Fingerprint Based Door Lock System

Abstract

This project presents a fingerprint-based door lock system using Arduino. It enhances security by allowing only authorized users to access the system through biometric authentication.

Introduction

Security is an important aspect in modern life. Traditional lock systems are not fully secure as keys can be lost or duplicated. Biometric systems like fingerprint recognition provide a more reliable solution.

Objective

The main objective is to design a secure, reliable, and low-cost door locking system using fingerprint technology.

Components Used

Arduino Uno, Fingerprint Sensor, Servo Motor/Relay, Buzzer, Jumper Wires, and Power Supply.

Working Principle

The system scans the fingerprint and compares it with stored data. If the fingerprint matches, the door unlocks. If not, access is denied and an alert is generated.

Advantages

High security, easy operation, no need for keys, and efficient performance.

Limitations

Requires proper fingerprint enrollment and may not work if the sensor is damaged or dirty.

Future Scope

Can be enhanced with IoT, mobile app control, and cloud storage for better performance.

Conclusion

The fingerprint door lock system provides a secure and modern solution for access control and can be widely used in smart homes.

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